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A study of the effect of bulges on bar formation in disc galaxies

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ABSTRACT

We use *N*-body simulations of bar formation in isolated galaxies to study the effect of bulge mass and bulge concentration on bar formation. Bars are global disc instabilities that evolve by transferring angular momentum from the inner to outer discs and to the dark matter halo. It is well known that a massive spherical component such as halo in a disc galaxy can make it bar stable. In this study, we explore the effect of another spherical component, the bulge, on bar formation in disc galaxies. In our models, we vary both the bulge mass and concentration. We have used two sets of models: one that has a dense bulge and high surface density disc, and the other model has a less concentrated bulge and a lighter disc. In both models, we vary the bulge to disc mass fraction from 0 to 0.7. Simulations of both the models show that there is an upper cut-off in bulge-to-disc mass ratio $M_{\rm b}/M_{\rm d}$ above which bars cannot form; the cut-off is smaller for denser bulges ($M_{\rm b}/M_{\rm d} = 0.2$) compared to less denser ones ($M_{\rm b}/M_{\rm d} = 0.5$). We define a new criterion for bar formation in terms of the ratio of bulge to total radial force ($F_{\rm b}/F_{\rm tot}$) at the disc scale lengths above which bars cannot form. We find that if $F_{\rm b}/F_{\rm tot} > 0.35$, a disc is stable and a bar cannot form. Our results indicate that early-type disc galaxies can still form strong bars in spite of having massive bulges.

Key words: methods: numerical – galaxies: bulges – galaxies: evolution – galaxies: kinematics and dynamics – galaxies: structure – dark matter.

1 INTRODUCTION

Nearly two-thirds of all disc galaxies in the observable universe are barred (Eskridge et al. 2000; Knapen, Shlosman & Peletier 2000; Menéndez-Delmestre et al. 2007; Barazza, Jogee & Marinova 2008; Hoyle et al. 2011). Many studies show that the fraction of barred galaxies decreases with redshift (Sheth et al. 2008; Nair & Abraham 2010; Melvin et al. 2014), but some observations have found constant bar fractions up to $z \sim 1$ (Jogee et al. 2004; Elmegreen, Elmegreen & Hirst 2004; Marinova & Jogee 2007) or even $z \sim 2$ (Simmons et al. 2014) that may be attributed to inclusion of small size bars and different galaxy selection criteria in different surveys. However, it is clear from these observations that most bars survive for at least $t \sim 8$ Gyr in galaxies in our low-redshift Universe.

In the Hubble sequence, bars vary from early- to late-type spiral galaxies (Elmegreen & Elmegreen 1985; Binney & Merrifield 1998; Buta et al. 2015). The bars associated with early-type, bulge-dominated galaxies (SBa, SBb) appear to be longer than those found in late-type spirals (Sc) (Erwin 2005). There is also a significant correlation between bar strength and bulge mass (Sheth et al. 2008; Skibba et al. 2012; Kim et al. 2014). The bulges themselves can be broadly classified into two types: the classical bulges that have a Sérsic index $n > 2$ and the more oval or discy pseudo-bulges that have $n < 2$ (Fisher & Drory 2008). Their structures suggests different formation mechanisms for each of them (Gadotti 2009). The origin of classical bulges is thought to be a result of major mergers (Kauffmann, White & Guiderdoni 1993; Baugh, Cole & Frenk 1996; Hopkins et al. 2009; Naab et al. 2014), multiple minor mergers (Bournaud, Jog & Combes 2007; Hopkins et al. 2010), monolithic collapse of primordial gas clouds (Eggen, Lynden-Bell & Sandage 1962), or the accretion of small satellites (Aguerri, Balcells & Peletier 2001). Pseudo-bulges are formed due to disc instability during secular evolution (Wyse, Gilmore & Franx 1997; Kormendy & Kennicutt 2004) or buckling instability of bars (Combes et al. 1990; Raha et al. 1991; Martinez-Valpuesta, Shlosman & Heller 2006), inward pull of gas by bars (Combes & Sanders 1981) and heating of bars by vertical resonances (Pfenniger & Norman 1990), resulting in boxy/peanut-shaped bulges.

Theoretical studies have shown that bars will form in self-gravitating, rotating discs when most of the kinetic energy is in rotational motion, i.e. in cold discs (Hohl 1971; Kalnajs 1972) unless there is a massive dark matter halo that stabilizes it against bar formation by providing a spherical gravitational field (Ostriker & Peebles 1973). However, recent studies (Athanassoula & Misiriotis 2002; Athanassoula 2003) show that a cold disc can become

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bar unstable despite the presence of a massive halo, which shakes the credibility of these previous studies. Global disc instabilities such as bar and spiral arms can also be triggered by disc perturbations due to interactions/mergers with minor satellites or larger galaxies (Noguchi 1987; Gerin, Combes & Athanassoula 1990; Barnes & Hernquist 1991; Salo 1991; Mayer & Wadsley 2004; Romano-Díaz et al. 2008). The evolution of bars in disc galaxies has been extensively studied with *N*-body simulations over the past few decades (Sellwood 1980; Combes & Sanders 1981; Efstathiou, Lake & Negroponte 1982; Athanassoula & Misiriotis 2002; Athanassoula 2002, 2003; Valenzuela & Klypin 2003; Machado, Athanassoula & Rodionov 2012; Saha & Naab 2013; Long, Shlosman & Heller 2014). The disc–halo interaction has been shown to play an important role not only in bar formation but also in bar rotation speeds (Debattista & Sellwood 2000; Weinberg & Katz 2007). Studies with live haloes show that bars evolve secularly in isolated galaxies through the exchange of angular momentum between disc stars and halo particles at radii corresponding to the disc resonances (Long et al. 2014). Faster rotating haloes may also trigger early bar formation (Long et al. 2014) but can also make bars weaker in the secular evolution phase. The bulges also gain angular momentum from the disc at resonances and this results in the spinning up of bulges (Saha, Martinez-Valpuesta & Gerhard 2012). These studies are similar to previous classical studies in which disc stars lose angular momentum at resonances (Lynden-Bell & Kalnajs 1972; Lynden-Bell 1979) or by being trapped in slowly rotating bar structures through dynamical friction (Tremaine & Weinberg 1984).

Although there have been many simulation studies of the effect of disc–halo interaction on bars, there are very few studies of the effect of bulges on bars. Since bulges contribute significantly to the radial force in discs, bulge masses and sizes can affect bar formation (Hohl 1976). One of the earlier theoretical studies of the effect of bulges on disc instabilities (Toomre 1981) showed that the presence of a strong bulge component in disc galaxies cuts off the feedback mechanism during swing amplification (Binney & Tremaine 2008) by introducing an Inner Lindblad Resonance (ILR) in the disc. The ILR does not allow trailing waves to go through the centre and emerge out as leading waves; this prohibits the growth of bar-type instabilities. However, detailed *N*-body simulations showed that this is not always true (Sellwood 1980); about 50–70 per cent of total galactic baryonic mass has to be in a spherical component to prevent the onset of bar instability in disc galaxies. These early studies used a spherical mass smaller than the disc but referred to it as a 'halo' rather than a bulge (Sellwood 1980). Further studies (Efstathiou et al. 1982) showed that bar-type instabilities can be sustained irrespective of the density of the bulge component. However, these studies used rigid bulges (Sellwood 1980; Efstathiou et al. 1982). When live bulges were used, it was found that the bulges gained angular momentum from the discs and this increased as the bar evolved (Sellwood 1980b). The disadvantage of all these early studies is that they did not include dark matter halo components in their galaxy models.

The above discussion holds for mainly isolated galaxies that are secularly evolving. However, when there are external perturbations to galaxy discs due to interactions with nearby galaxies, stellar bars can form even in the presence of a strong bulge. In such cases, the ILR is not able to prevent the growth of bars. Later studies that included halo components in their models suggested that the presence of massive bulges make discs stable against bar formation (Sellwood & Evans 2001). It is well known that the probability of having an ILR is higher when there is a massive bulge as the angular velocity Ω of stars in the disc increases and hence the critical value $\Omega_{\rm crtical} = \Omega_p + \kappa/2$ will be reached, therefore cutting off feedback mechanism as discussed in the last paragraph, where Ω_p and κ are the pattern speed of bar and epicyclic frequency of stars. However, there are many non-linear processes in galaxy evolution that allow bar formation in bulge-dominated disc galaxies (Widrow, Pym & Dubinski 2008; Dubinski, Berentzen & Shlosman 2009). Cosmological hydrodynamic simulations (Scannapieco & Athanassoula 2012) also show that strong bars can form in galaxies that have prominent bulges rather than those without bulges.

Recent studies have shown that bars can also affect bulges by increasing their spin (Saha, Martinez-Valpuesta & Gerhard 2012; Saha 2015; Saha, Gerhard & Martinez-Valpuesta 2016). However, there is no detailed study that explores the effect of bulge masses on bar formation and its evolution. In this paper, we re-visit the dependence of bar formation on bulges but with a difference; we vary not only the bulge mass but also its concentration. We also focus on the angular momentum transfer between the disc and bulge, since angular momentum transfer between discs and dark matter haloes has been found to play a key role in bar evolution (Long et al. 2014). Thus, the goal of this paper is to determine how bulge mass and concentration affects bar formation, morphology, pattern speed, and angular momentum transfer between the different galaxy components (bulge, disc, halo). Our models start from bulgeless galaxies to bulge-dominated discs, where the bulge-to-disc mass ratio can be as high as 70 per cent. We have two sets of models, one with a concentrated bulge and the other with a less concentrated bulge. The plan of the paper is as follows. In Section 2, we discuss the numerical techniques used for this work that includes initial condition generation and simulation methods. In Section 3, we discuss the evolution of various bar parameters in the different cases, such as pattern speed, bar strength, angular momentum transfer between disc and bulge components and the origin of pseudo-bulges in our models. Apart from this in the same section we discuss a model independent parameter that can be used to determine the limiting bulge force for bar formation. In Section 4, we discuss the implication of our results for bulge–bar correlations in disc galaxies. Finally, in the Section 5, we summarize our work.

2 NUMERICAL TECHNIQUE

2.1 Initial conditions of modelled galaxies

For generating initial disc galaxy models, we have used the code GALIC (Yurin & Springel 2014) that iteratively populates orbits with given density distribution using the parts of Schwarzschild technique. This code finds steady-state solution for the collisionless Boltzmann equation by iteratively adjusting the velocity of disc stars in a given density distribution to generate equilibrium galaxy models. The number of particles used for this simulation is 10^6 dark matter halo particles, 10^5 disc particles, and 5×10^4 bulge particles.

We present here two categories of galaxy models that we denote as MA and MB (here onwards in this article) that have disc, bulge, and dark matter halo components. The MA models have relatively smaller discs and higher disc mass surface densities than that of MB models. The purpose of taking these two models is to see the growth and evolution of bar instabilities in both larger and smaller disc galaxies. The galaxy MA models have more concentrated or denser bulges compared to MB models. The details of various parameters in initial models generated is shown in Table 1. From top to bottom, the bulge mass content is increasing in both MA and MB models, respectively. Figs 1 and 2 show initial and final rotation curves, initial surface density, and initial Toomre's parameter with

Table 1. Initial disc models with increasing bulge masses.

Models	$\frac{M_B}{M_D}$	$\frac{M_B}{M_T}$	$\frac{M_D}{M_T}$	$\frac{M_H}{M_T}$	$\frac{R_b}{R_d}$	$Q(R_D)$	M_B (10^{10}M$_\odot$)	t_{OP}
MA00	0	0	0.1	0.9	0	1.077	0	0.180
MA01	0.1	0.01	0.1	0.89	0.174	1.123	0.64	0.179
MA02	0.2	0.02	0.1	0.88	0.180	1.171	1.28	0.179
MA03	0.3	0.03	0.1	0.87	0.186	1.220	1.93	0.177
MA04	0.4	0.04	0.1	0.86	0.192	1.269	2.55	0.175
MA05	0.5	0.05	0.1	0.85	0.198	1.319	3.19	0.176
MA06	0.6	0.06	0.1	0.84	0.204	1.368	3.83	0.177
MA07	0.7	0.07	0.1	0.83	0.210	1.418	4.47	0.177
MB00	0	0	0.1	0.9	0	1.239	0	0.192
MB01	0.1	0.01	0.1	0.89	0.439	1.273	1.86	0.193
MB02	0.2	0.02	0.1	0.88	0.447	1.305	3.72	0.194
MB03	0.3	0.03	0.1	0.87	0.456	1.337	5.58	0.195
MB04	0.4	0.04	0.1	0.86	0.465	1.370	7.44	0.196
MB05	0.5	0.05	0.1	0.85	0.473	1.401	9.30	0.197
MB06	0.6	0.06	0.1	0.84	0.481	1.433	11.16	0.198
MB07	0.7	0.07	0.1	0.83	0.490	1.465	13.02	0.199

Note. Column 1: model name; column 2: ratio of bulge to disc mass; column 3 ratio of bulge to total galaxy mass; column 4: ratio of disc to total galaxy mass; column 5: ratio of halo to total mass; column 6: ratio of half-mass bulge radius to disc scale length (R_b/R_d); column 7: Toomre's parameter at disc scale length R_d; column 8: bulge mass; column 9: Ostriker and Peebles criterion for bar instability t_{OP} explained in Section 3.4.

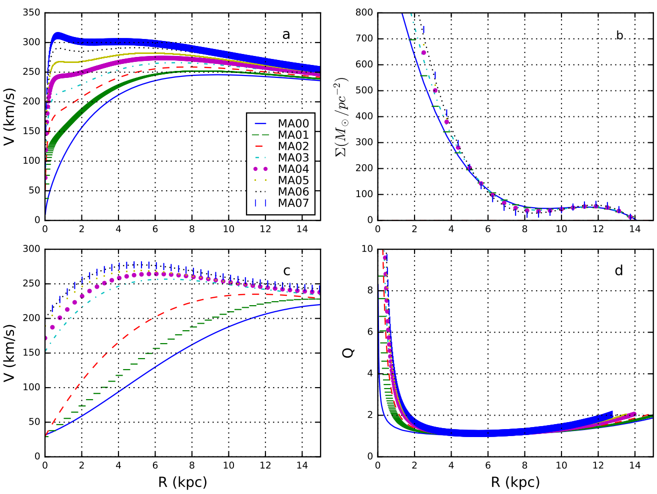

Figure 1. (a) Initial rotation curves of disc stars; (b) surface density; (c) final rotation curve at 9.78 Gyr; (d) Toomre's parameter variation with radius for all the MA models.

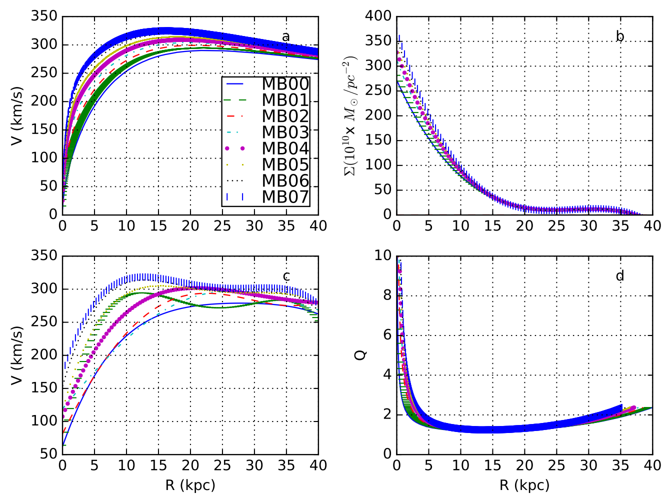

Figure 2. (a) Initial rotation curves of disc stars; (b) surface density; (c) final rotation curve at 9.78Gyr; (d) Toomre's parameter variation with radius for all the MB models.

radius for both MA and MB models, respectively. Figs 3 and 4 show the contribution of the different galaxy components to the rotation curves of the galaxy models. Figs 5 and 6 show the initial radial velocity dispersion of stellar disc for models MA and MB, respectively.

The density profile for a spherically symmetric halo is chosen as

$$\rho_h = \frac{M_{\rm dm}}{2\pi}\frac{a}{r(r+a)^3}, \tag{1}$$

where a is the scale length of the halo component. This scale length is related to the concentration parameter of the NFW halo having $M_{\rm dm} = M_{200}$ (Springel, Di Matteo & Hernquist 2005) so that the inner shape of the halo is identical to the NFW halo. Here, a and c are related as following:

$$a = \frac{R_{200}}{c}\sqrt{2[ln(1+c) - c/1+c]}, \tag{2}$$

where M_{200} and R_{200} are virial mass and virial radius for NFW halo, respectively.

The density profile for the disc component has an exponential distribution in the radial direction and sec h^2 profile in the vertical direction.

$$\rho_d = \frac{M_d}{4\pi z_0 h^2}\exp\left(-\frac{R}{R_{\rm d}}\right)\sec h^2\left(\frac{z}{z_0}\right), \tag{3}$$

where $R_{\rm d}$ and z_0 are the radial scale length and vertical scale length, respectively. The values of these parameters are listed in Table 1 for all of our models.

Finally, the bulge component in our models has a Hernquist density profile given by

$$\rho_b = \frac{M_{\rm b}}{2\pi}\frac{R_{\rm b}}{r(r+R_{\rm b})^3}, \tag{4}$$

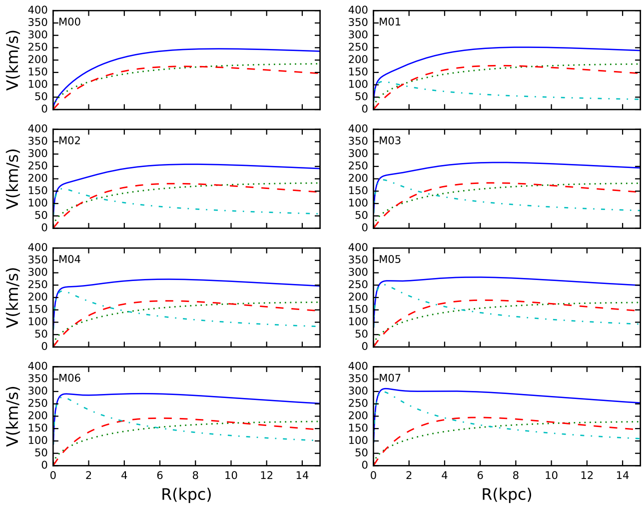

Figure 3. Represents rotational curves for all of our MA models. Here, solid line shows rotational velocity due to all the components; halo is shown by dotted line; disc is shown by dashed line; and bulge is shown by dash–dotted line.

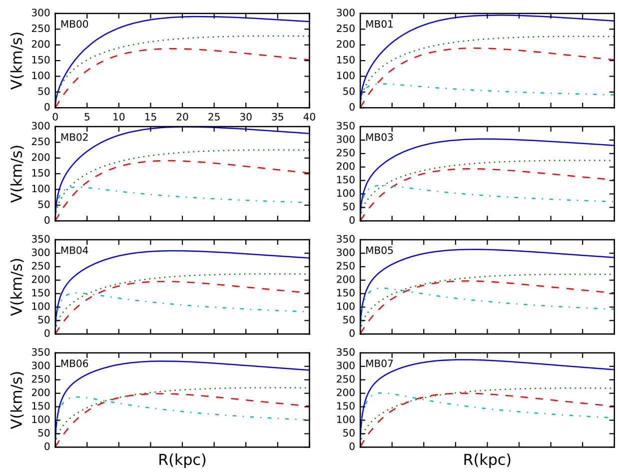

Figure 4. Represents rotational curves for all of our MB models. Here, solid line shows rotational velocity due to all the components; dotted line shows halo; dashed line shows disc ; dash–dotted line shows bulge..

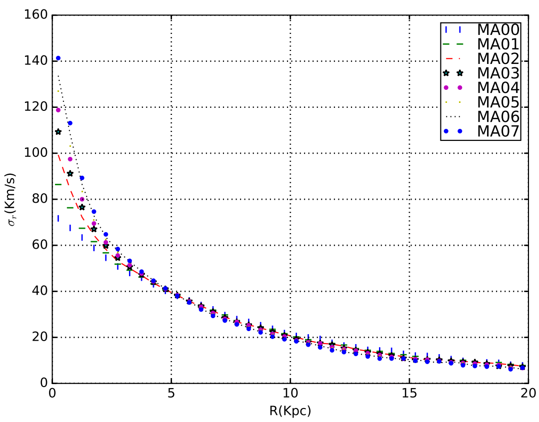

Figure 5. Initial radial velocity dispersions for all the MA models.

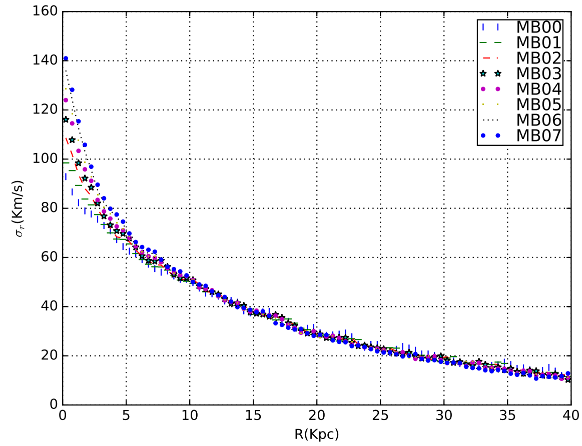

Figure 6. Initial radial velocity dispersions for all the MB models.

where M_{b} and R_{b} are total bulge mass and bulge scale length, respectively. These values are listed in Table 1.

Table 1 also contains many other parameters like bulge to disc fraction, bulge to total galaxy mass fraction, disc to total galaxy mass fraction, halo to total galaxy mass fraction, Toomre's factor at disc scale length (R_{d}), and bulge mass. For all the MA and MB-type models, the total mass of the galaxies is decided by the velocity at virial radius which in our model is equal to 140 and 200 km s^{-1}, respectively. Total galaxy mass for MA galaxy model is equal to $63.8 \times 10^{10}\mathrm{M}_{\odot}$, and for MB-type models, it is $186 \times 10^{10}\mathrm{M}_{\odot}$. Mass contents in disc components for all the MA and MB models are $6.38 \times 10^{10}\mathrm{M}_{\odot}$ and $18.6 \times 10^{10}\mathrm{M}_{\odot}$, respectively. In our models, ratio of particle masses in halo, disc, and bulge varies for different models. For MA models, ratio of particle masses goes from 1:1.12:0.22 for MA01 to 1:1.2:1.68 for MA07. For MB models, ratio of particle masses of halo, disc, and bulge varies from 1:1.2:0.22 for MB01 to 1:1.20:1.03 for MB07. Both of our models are dark matter dominated, and dark matter makes up 90 per cent of the total galaxy mass. In our models, we have chosen the spin parameter for the dark matter halo component to be 0.035 that is shown to be the most probable value from cosmological simulations (Bullock et al. 2001). Bulges in all the models are non-rotating. We have checked that all the models are locally stable as Toomre's parameter is greater than 1 for all the models throughout the disc. The Toomre's factor varies with radius and is given by $Q(r) = \frac{\sigma(r)\kappa(r)}{3.36G\Sigma(r)}$. Here, $\sigma(r)$ is the radial dispersion of disc stars, $\kappa(r)$ is the epicyclic frequency of stars, and $\Sigma(r)$ is the mass surface density of the disc.

2.2 Simulation method

After all, the initial galaxy models are prepared we evolve these models in isolation with GADGET-2 code (Springel 2005). We evolved these galaxies up to 9.78 Gyr to check secular evolution of the discs in our models. This code uses various types of leapfrog methods for time integration. It uses the Tree method (Barnes & Hut 1986) to compute gravitational forces between different particles. The opening angle for the tree is chosen as θ_{tot} =0.4. The softening length for halo, disc, and bulge components has been chosen 30, 25, and 10 pc, respectively. We mention our results in terms of code units. Both GALIC and GADGET-2 codes have unit mass equal to 10^{10} $\mathrm{M}_{\odot}$, unit distance is 1 kpc, unit velocity is 1 km s^{-1}.

The introduction of concentrated bulges increases the frequency of two-body interactions in our simulations and affects the conservation of angular momentum of the galaxy. This means that the force calculation accuracy plays an important role in determining positions and velocities of the particles and therefore angular momentum conservation. We have done many test simulations to determine the most important parameters for conserving the angular momentum. We found that reducing the softening length does not help much in conserving the angular momentum. Instead for our current simulation, reducing the time step of integration (η) and the force accuracy parameter over the evolution time period are the most important parameters for angular momentum conservation (Klypin et al. 2009). We haves used the values $\eta \leq$0.15 and force accuracy parameter $\leq$0.0005 in most of the simulations. As a result in all of our models, the angular momentum is conserved to within 1 per cent of the initial value.

2.3 Bar strength and pattern speed

Bar strength has been defined in different ways in the literature (Combes & Sanders 1981; Athanassoula 2003). In our study for defining bar strength, we have used the mass contribution of disc stars to the $m = 2$ Fourier mode:

$$a_2(R) = \sum_{i=1}^{N} m_i \cos(2\theta_i) \qquad b_2(R) = \sum_{i=1}^{N} m_i \sin(2\theta_i), \tag{5}$$

where a_2 and b_2 are defined in the annulus around the radius R in the disc, m_i is mass of ith star, and θ_i is azimuthal angle. We have defined the bar strength as

$$\frac{A_2}{A_0} = \max\left(\frac{\sqrt{a_2^2 + b_2^2}}{\sum_{i=1}^{N} m_i}\right). \tag{6}$$

We calculate the pattern speed(Ω_B) of the bar by measuring change in phase angle $\phi = \frac{1}{2}\tan^{-1}\left(\frac{b_2}{a_2}\right)$ of the bar which is calculated using the Fourier component in the annulus corresponding to maximum bar strength. We use annular regions of size 1 kpc for disc particles only.

2.4 Angular momentum calculation

We measured the angular momentum of the different components of a galaxy separately; disc, bulge, and halo. Angular momentum of a particle is calculated using the product of its particle mass, radial distance from galactic centre, and circular velocity. In this paper, we plot time evolution of total angular momentum of each component in the galaxy.

3 RESULTS

3.1 Evolution of bar strength

We tracked bar formation and evolution up to 9.78 Gyr in all of the models from MA00 to MA07. Fig. 7 shows the evolution of bar strength for these models. The X–Y face-on view of all the MA models after 9.78 Gyr is shown in Fig. 8. We see that the model without bulge (MA00) grows to peak bar strength around 1.5 Gyr during dynamical evolution and its bar strength decreases rapidly before secular evolution phase. In secular evolution, its bar strength increases gradually with time. In the model with low-mass bulges (MA01), the bar grows to peak strength around 3 Gyr and sustains its strength during secular evolution phase. Further with increasing bulge masses in model MA02, bar instability sets in later and reaches

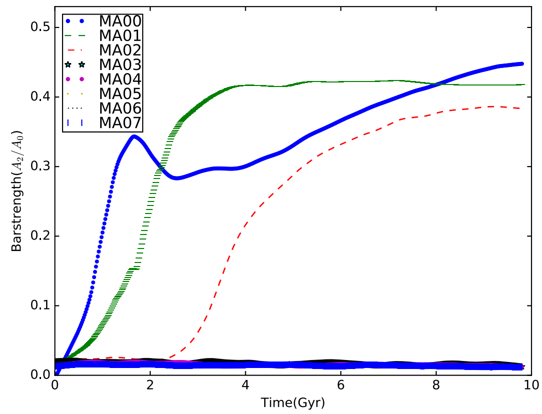

Figure 7. Evolution of bar strength with time for all MA models.

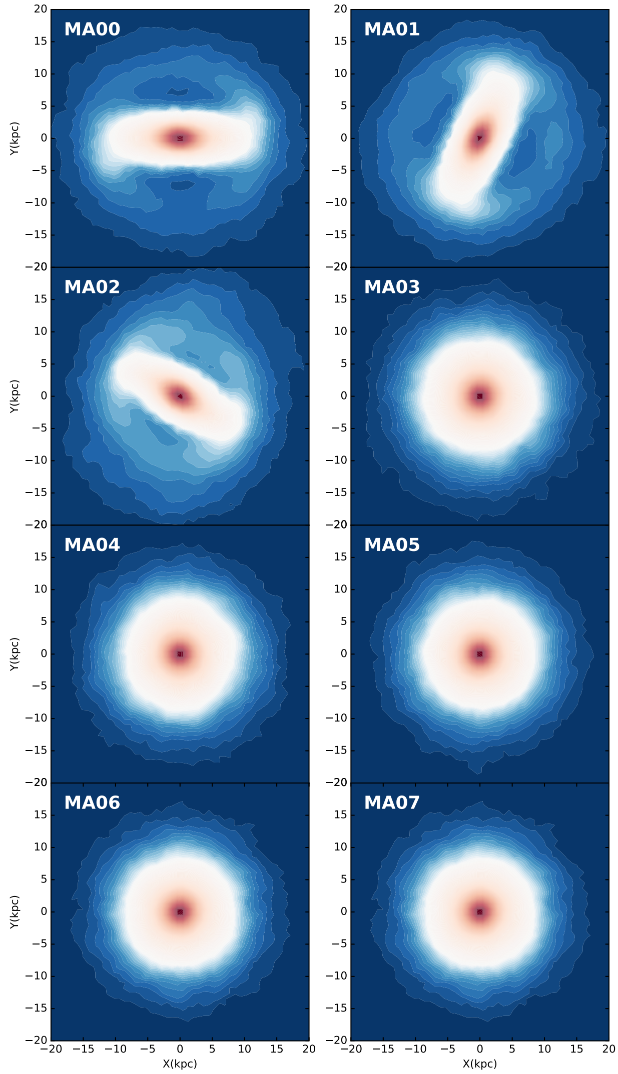

Figure 8. X–Y cross-section of all the MA models at 9.78 Gyr.

peak value around 7 Gyr. In the later models, MA03, MA04, MA05, MA06, and MA07 that have bulge-to-disc fraction > 0.3 bar type of instability is suppressed completely due to the presence of massive bulge.

Fig. 9 shows evolution of bar strength for the models MB00 to MB07, which have less concentrated bulges and lower disc surface

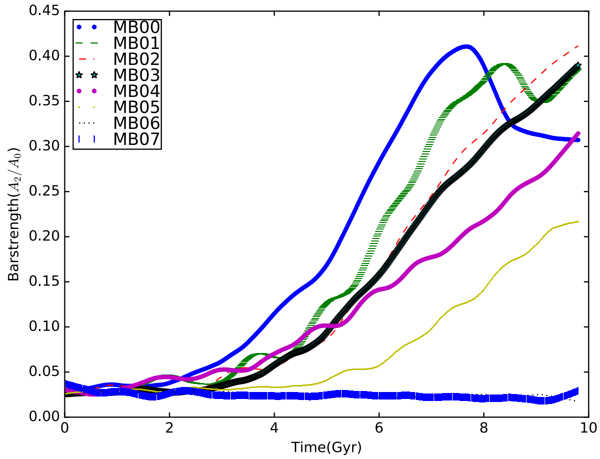

Figure 9. Evolution of bar strengths with time for all MB models.

densities compared to models MA00 to MA07. As in the previous models, the simulation is run for 9.78 Gyr. The main difference in the bar evolution of models MA and MB is that the bar forms much later in model MB. This is due to the lower mass surface density Σ of the disc that leads to a lower disc self-gravity and hence instabilities take longer to develop. We find that the model with no bulge (MB00) shows bar-type instability, for which bar strength peaks at around 7.5 Gyr and the bar gets weaker with further evolution. On introduction of a bulge in our MB models, we see that the bar strength shows non-linear trends as a function of bulge-to-disc fraction. For small bulge-to-disc ratios of 0.1–0.3, the bar triggering time-scale increases with bulge mass and the peak bar strength remains almost the same. As we increase bulge fraction from 0.4 to 0.6, we see that the bar formation time-scale increases and peak strength reduces for the time we evolve our model. For models with bulge-to-disc fraction 0.7, a bar does not form at all. The *X*–*Y* face-on views of all the MB models after 9.78 Gyr is shown in Fig. 10.

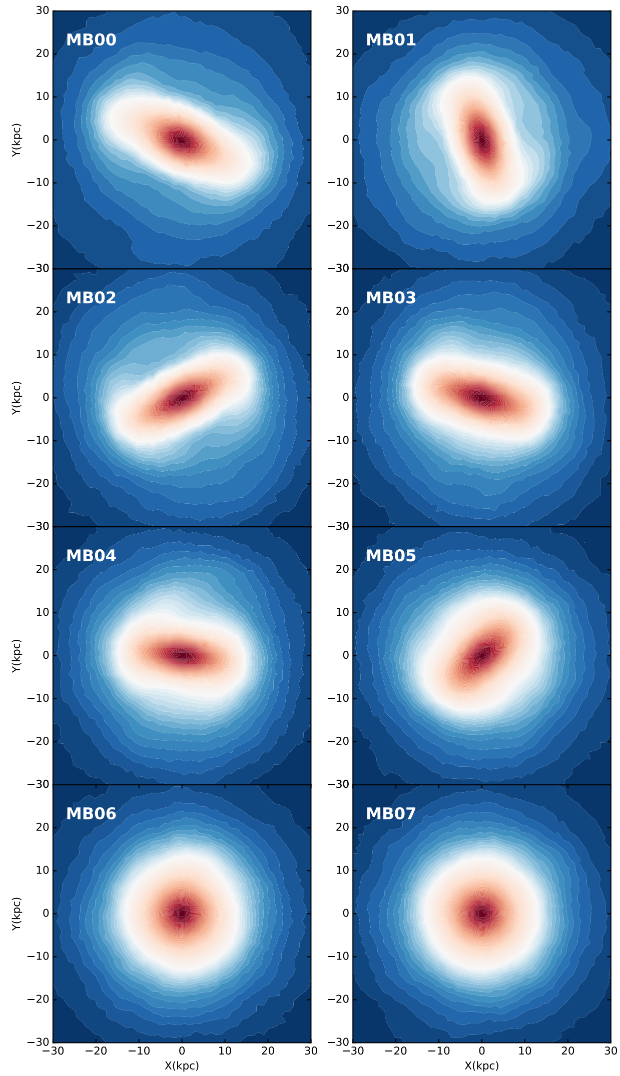

Figure 10. *X*–*Y* cross-sectionof all the MB models at 9.78 Gyr.

3.2 Evolution of bar pattern speed

We show the evolution of bar pattern speed (Ω_p) with time in Fig. 11 for all MA models that form bars. In this plot, we do not have models MA03 to MA07 because their discs do not form bars. It is very clear that the discs that have more massive bulges have bars with higher Ω_p. Also, in all the models, the Ω_p decreases with time and the rate of decrease in pattern speed increases with increase in bulge mass fraction.

The variation of bar pattern speed with time for all MB models which form bars is shown in Fig. 12. Note that we can plot Ω_p only after the bars form and start growing. So this plot excludes models MB06 and MB07 that do not form bars. As in models MA, as the bulge fraction increases (from MB00 to MB05), the Ω_p value is higher. Thus, bar rotation is faster for bulge-dominated galaxies, whatever be the bulge concentration or disc surface density. This is because the increase in bulge mass makes the inner disc potential deeper resulting in larger angular velocities. In all the bar-forming models, Ω_p shows little variation for several Gyr until 8–9 Gyr, after which Ω_p decreases sharply with time. We interpret that this may be related to the thickening of the bar as the bar growth peaks after this time period.

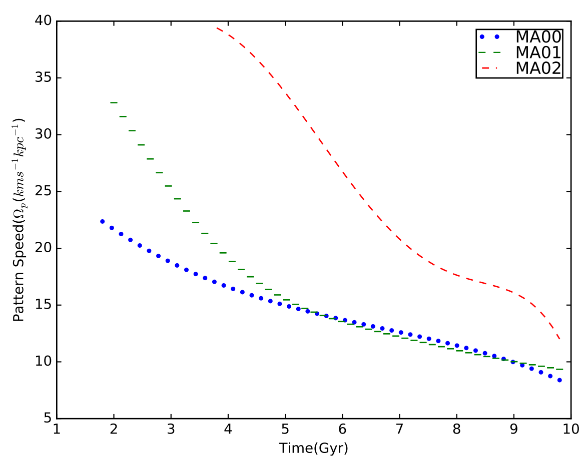

Figure 11. Pattern speed evolution with time of all bar-forming MA models.

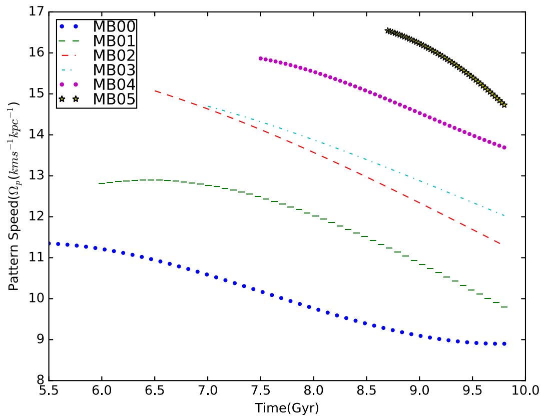

Figure 12. Pattern speed evolution with time of all bar-forming MB models.

3.3 Angular momentum exchange

Several studies have shown that angular momentum exchange between the disc and halo plays a vital role in the formation and evolution of bars (Athanassoula 2003; Valenzuela & Klypin 2003; Martinez-Valpuesta et al. 2006; Saha & Naab 2013; Long et al. 2014). We see this in both MA- and MB-type models in our simulations. In this sub-section, we discuss the importance of angular momentum transport between the disc, halo, and bulge in our models. We have also plotted the total angular momentum exchange with time for the individual galaxy components in Figs 13 and 14 for MA- and MB-type models, respectively.

3.3.1 MA models: high density bulges and discs

The overall change in angular momentum for bulge, disc and halo components of MA models are shown in Fig. 13. We see that disc component loses total angular momentum by huge amount in bar-forming models compared to models that do not show bar instability. We also see that angular momentum loss rate for the discs increases sharply when the bars reach their peak strength; the time-scale for which increases with increase in bulge mass. We can see that the total angular momentum gain is maximum for the halo component and is around 15–20 times that of the bulge component.

We also note that the total angular momentum of the bulge component first increases with increase in bulge mass for models MA00, MA01, and MA02 that show bar instabilities. Further increase in bulge mass in model MA03 leads to decrease in bulge angular momentum. After this, an increase in bulge mass (model MA04 onwards) results in a slow increase in the bulge total angular for all the bulges and could represent the slow spin up of bulges.

For the halo component, the total angular momentum exchange is maximum for the MA00 model that does not have a bulge. On introduction of bulge for models MA01 and MA02, the halo component gains angular momentum after the bar gains its peak strength around 2 and 4 Gyr, respectively. For models MA03 onwards where the bar instability is not triggered and the bulges mass increases, the angular momentum exchange becomes very small.

3.3.2 MB models: low-density bulges and discs

In all the MB models, the bar forms after 6 Gyr that is much later than in the MA models, mainly because the disc mass surface density is much lower. Fig. 14 shows the total angular momentum change for bulge, disc, and halo components of all the MB models. We see that disc component loses angular momentum by large amount in bar-forming models. Similar to MA models, here also the rate of decrease in angular momentum correlates with the bar strength peaking time-scale that is shown in Fig. 11. Total loss of angular momentum by disc component decreases with increase in bulge mass as we go from MB01 to MB07. In these models, we see that both halo and bulge components gain angular momentum. Also, the total angular momentum gain is maximum for the halo component and is around 5–40 times that of the bulge component.

We see that increase in total angular momentum of bulge increases with increase in bulge masses. It can also be seen clearly that rate of increase of angular momentum for bulge components increase after bar strengths peak that varies in all the models.

The rate of gain in angular momentum by the halo component also start increasing only after the bar gets its peak strength. Further total change in angular momentum of halo component decreases with increase in bulge mass. This clearly shows that a massive bulge component delays bar formation by curbing angular momentum transport from disc component to halo component.

3.4 Bar instability criterion

Earlier works have shown that cold stellar discs (Hohl 1971; Kalnajs 1972; Athanassoula 2003) with low velocity dispersion can become bar unstable within a few gigayears of evolution. Ostriker and Peebles have given a criterion that if the ratio of rotational kinetic energy to potential energy of a disc exceeds 0.14 ($t_{\rm OP} > 0.14$), then the galaxy disc is bar unstable (Ostriker & Peebles 1973). In all of our models (MA and MB), we see that all of the $t_{\rm OP} > 0.14$. This suggests that all of our models should have formed bars as the disc is cold enough to become bar unstable. We show the values of $t_{\rm OP}$ for all the models in Table 1. We find that MA models become bar unstable when $t_{\rm OP} > 0.177$ and MB models become unstable when $t_{\rm OP} > 0.197$. Thus, the $t_{\rm OP}$ does not appear to be constant in the different disc models and depends on the disc surface mass density as well as the bulge mass concentration. We suggest that this shift in Ostriker and Peebles criterion to higher values is due to live/spinning nature of halo (Saha & Naab 2013) and bulge components.

Toomre (1981) has shown that bar instabilities can occur through the feedback mechanism during swing amplification. This feedback mechanism is cut down in the presence of ILRs that do not allow the waves to propagate and hence bars should not form in the presence of ILRs. However, we do not find that the presence of ILRs in our simulation is a deciding criterion for bar formation.

As we know bulges are the prominent mass component in the centres of galaxies. Hence, they contribute significantly to disc rotation in the central regions. We have found that all the bar-forming models have fractional bulge to total radial force values, $F_{\rm b}/F_{\rm tot}$, at the disc scale radius $R_{\rm d}$, to have values of less than 0.35. Here, $F_{\rm b} = V_b^2/R_{\rm d}$ and $F_{\rm tot} = V_{\rm tot}^2/R_{\rm d}$, where $V_{\rm b}$ is velocity due to bulge component and $V_{\rm tot}$ is velocity due to bulge, disc, and halo components. We have plotted this ratio $F_{\rm b}/F_{\rm tot}$ with radius in Figs 15 and 16 for the models MA and MB. The cut-off for fractional bulge force is shown clearly in Fig. 17. The dots of different shapes represent the values of $F_{\rm b}/F_{\rm tot}$ at disc scale lengths and the value is never larger than 0.35. Hence, our work shows that bars can form in disc galaxies only when $F_{\rm b}/F_{\rm tot} < 0.35$. This happens because velocity dispersion of disc stars increases as result of increase in radial force due to increasing mass of bulge component. We thus propose that this is a new criterion for bar

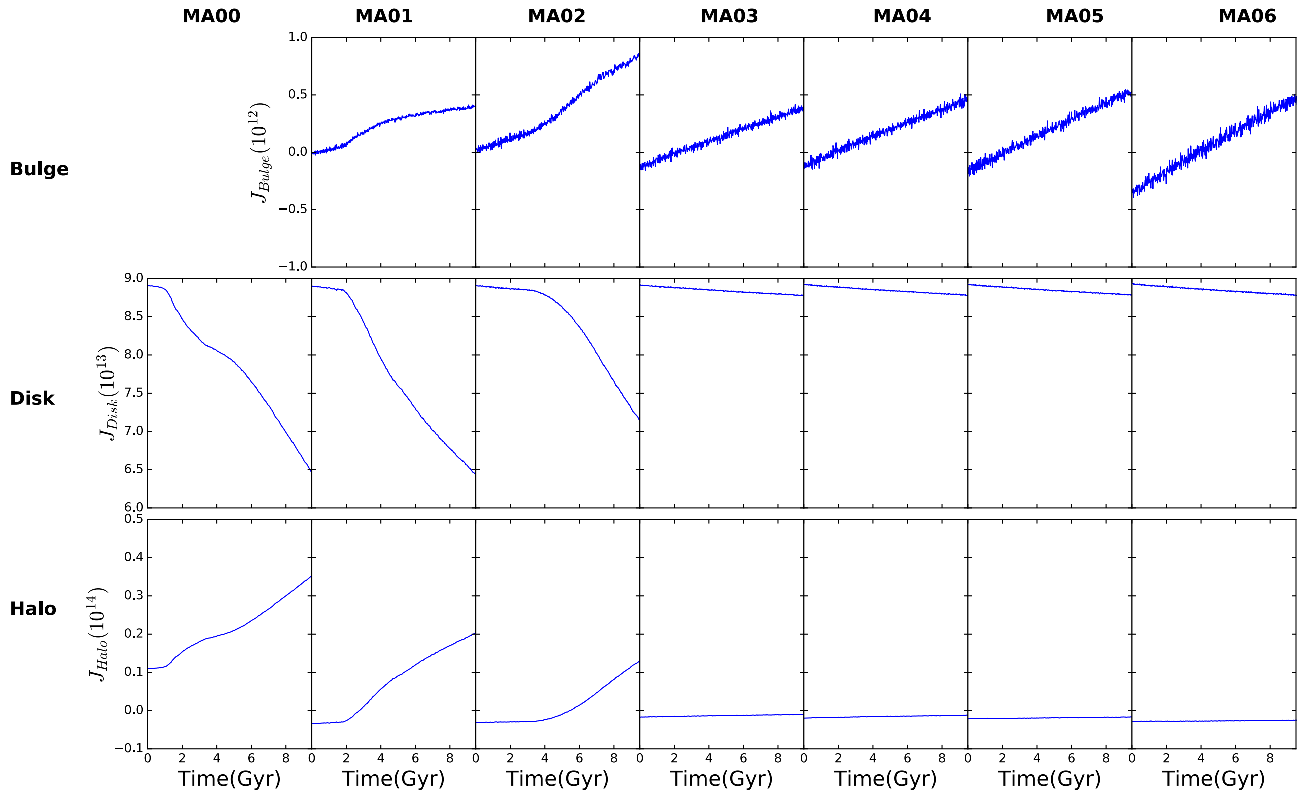

Figure 13. Total angular momentum for all the components (bulge, disc, and halo) of all MA models.

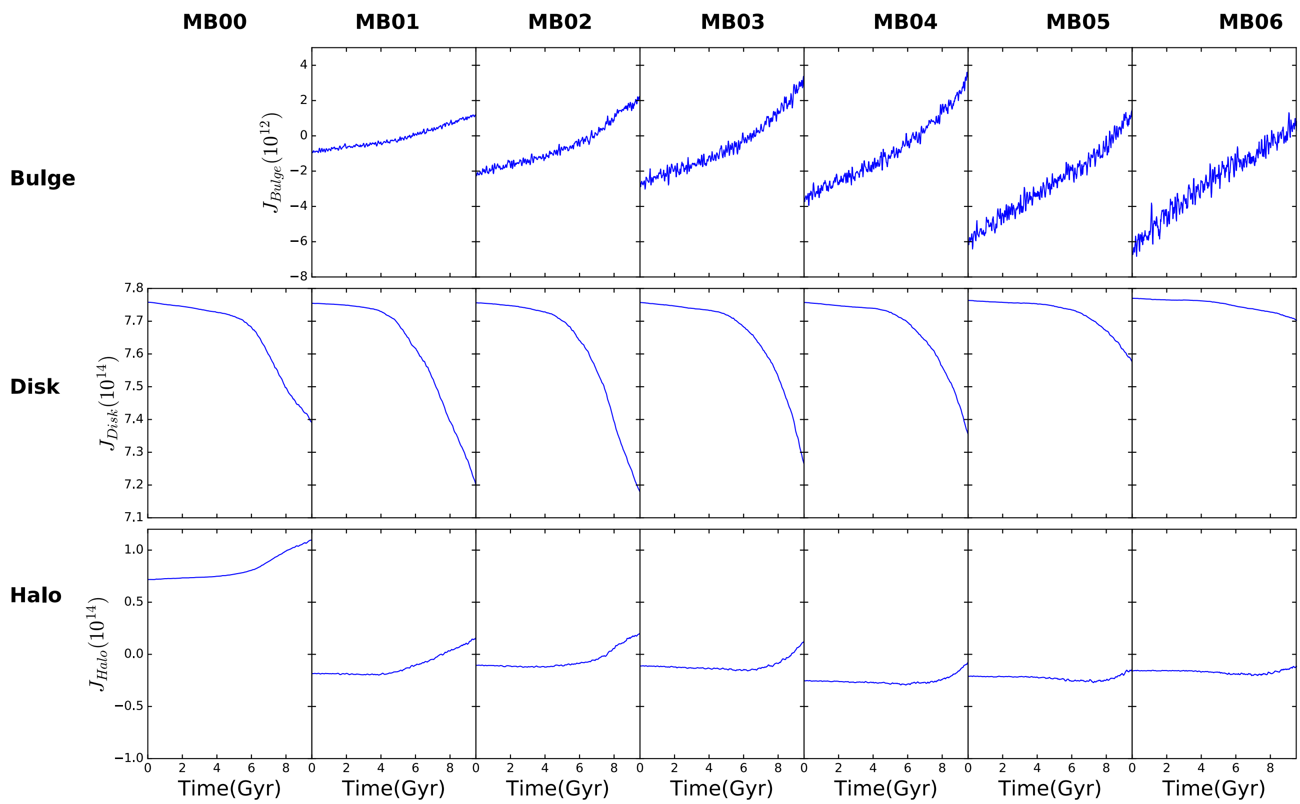

Figure 14. Total angular momentum for all the components (bulge, disc, and halo) of all MB models.

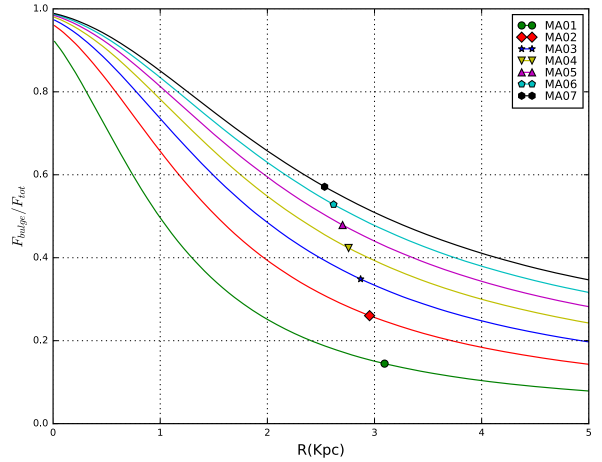

Figure 15. Fractional bulge force at disc scale length for MA models (various shape dots).

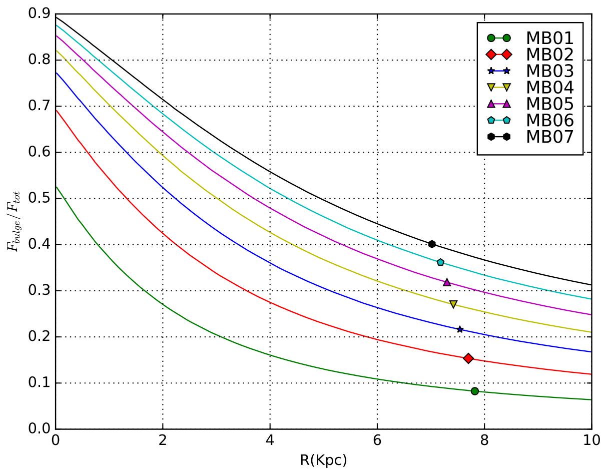

Figure 16. Fractional bulge force at disc scale length for MB models (various shape dots).

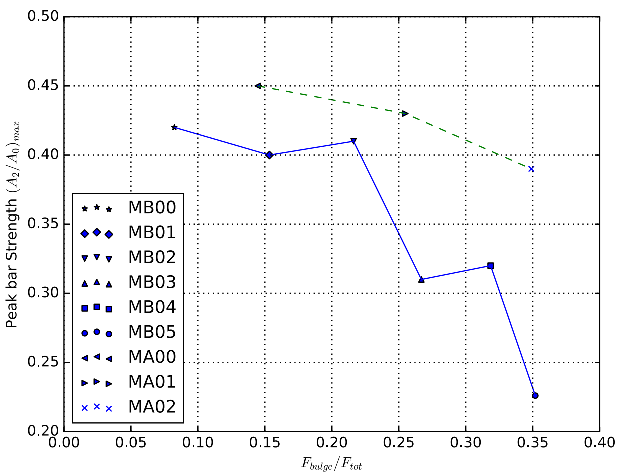

Figure 17. Peak bar strength with respect to fractional bulge force for all the bar-forming MA (dashed line) and MB models (solid line).

formation in terms of radial force due to the bulge mass in the disc. This criterion gives clear idea about the amount of bulge force that prevents bar formation, regardless of the bulge mass or concentration.

3.5 Boxy bulges in the models

Figs 18 and 19 show the peanut/box(P/B) shape features at the end of evolution in both the MA and MB models, respectively, in the y–z planes at both start and end of the simulations. We see that only the bar-forming models show P/B feature. Their origin is due to heating of bar through vertical resonances (Pfenniger & Norman 1990). P/B features are stronger for MA models than MB models. This is because the bar forms earlier ($t\sim$ 2 Gyr) in MA models, and this provides sufficient time for bar heating during the secular evolution phase. While in MB models the bar forms much later ($t\sim 6$ Gyr) and it does not have enough time to secularly evolve. We find that there is a weak correlation of these P/B feature and the classical bulge mass fraction in both the MA and MB models.

4 IMPLICATIONS OF OUR RESULTS

Our simulations show that the formation of bar instabilities changes with bulge mass and bulge concentration. Both effects can be explained by the criterion $F_{\rm b}/F_{\rm tot} < 0.35$ for bar formation, as described in the previous section. The cut-off implies that angular momentum transfer amongst galaxy components gets damped due to increase in velocity dispersion of stars due to introduction of massive bulges. This means that galaxies with strong bulges will not form bars easily in their discs, perhaps only during interactions with other galaxies as well. Or in other words, strong bulges make discs extremely stable. Such strong bulges may form during galaxy formation epochs, in which case the galaxies do not form bars easily or during secular evolution when gas is driven into the nuclear region by bars or spiral arms, leading to the build-up of central mass concentrations (Norman, Sellwood & Hasan 1996; Das et al. 2003, 2008).

Our bulge-related bar instability criterion can be easily related to observations in the following way. For a given galaxy image in the near-infrared (which is a waveband that traces the main stellar content of galaxy), a bulge–disc–bar image decomposition will yield the luminosity of the individual components ($L_{\rm bulge}$, $L_{\rm disc}$, $L_{\rm bar}$). Further force due to bulge component can be calculated through mass of bulge ($M_{\rm bulge}$), which can be obtained by multiplying M/L ratio with $L_{\rm bulge}$. Rotation curve ($V_{\rm tot}$) and disc scale length of the galaxy can be used to obtain total force due to galaxy at disc scale length. Hence, our criterion can be written in observable quantities as

$$\frac{F_{\rm b}}{F_{\rm tot}} = \frac{G M_{\rm bulge}}{R_{\rm d} V_{\rm tot}^2}. \quad (7)$$

This effect of strong bulges can be clearly seen in observations of bulge-dominated disc galaxies. Bars are not common in S0 galaxies that have classical bulges and bulge–disc decomposition for S0s reveal a bulge to total luminosity (B/T) value equal to 0.35 (Barway et al. 2016). This value is similar in nature to our criterion of $F_{\rm b}/F_{\rm tot} < 0.35$. Studies of late-type spiral galaxies indicate that bulge-dominated spirals have a smaller fraction of barred galaxies compared to disc-dominated ones (Barazza, Jogee & Marinova 2008). However, later studies with Galaxy Zoo have shown that bars tend to form in discs with stronger bulges (Hoyle et al. 2011).

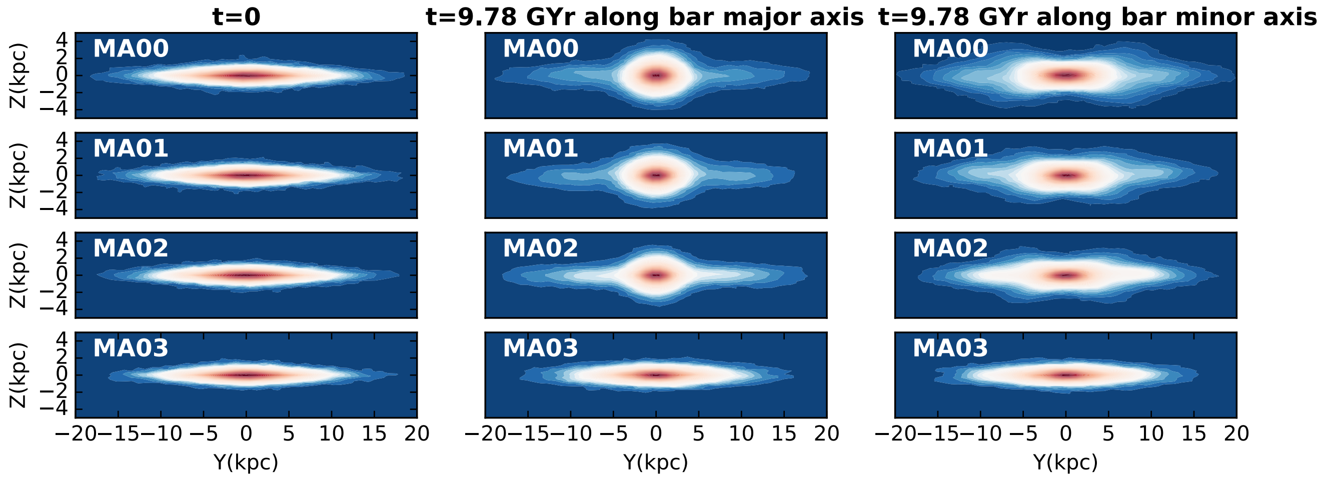

Figure 18. First column shows edge-on view (Y–Z cross-section) of MA models at t = 0; second and third columns show two different edge-on views at t = 9.78 Gyr.

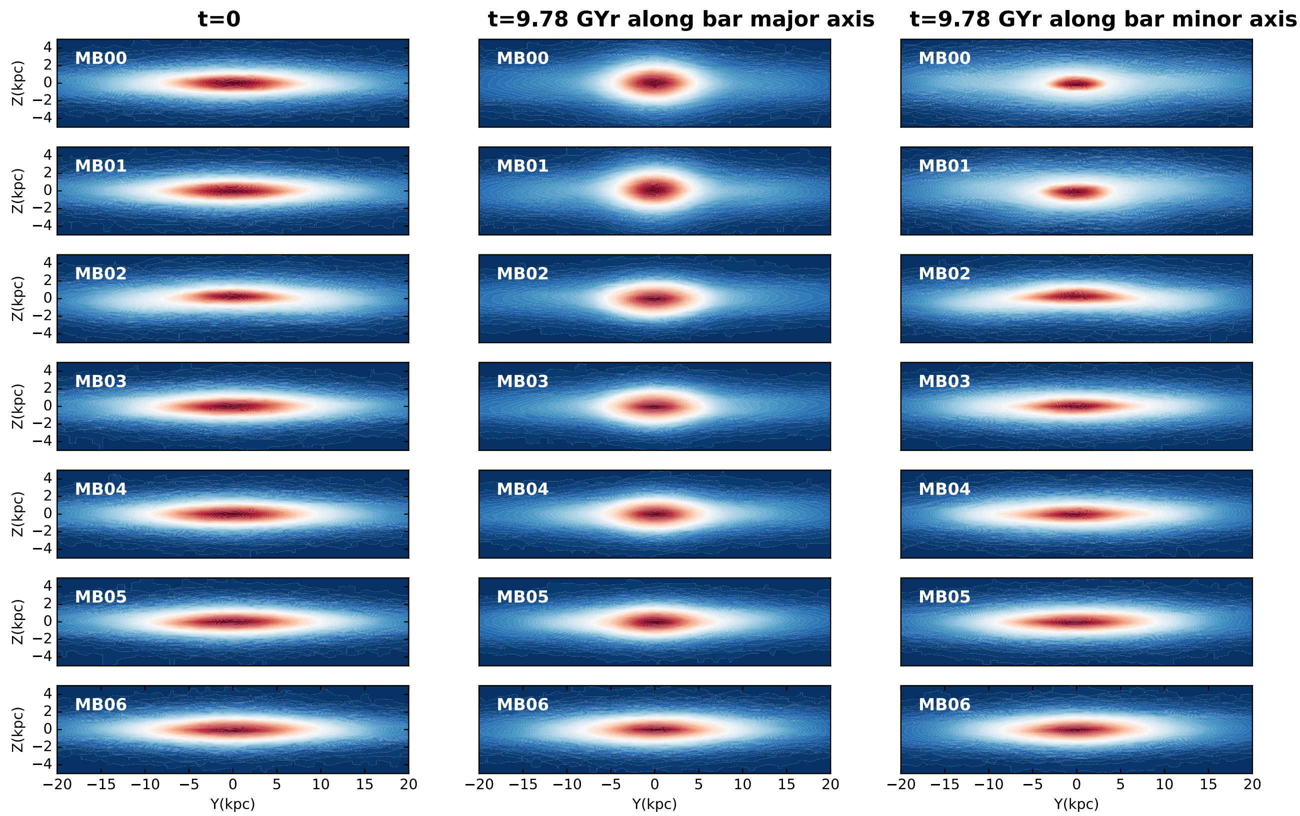

Figure 19. First column shows edge-on view (Y–Z cross-section) of MB models at t = 0; second and third columns show two different edge-on views at t = 9.78 Gyr.

A more rigorous study of B/T values for a large sample of barred galaxies is required to fully test our criterion. Another important implication of our study is that bars do form in dark-matter-dominated discs. Our MB models are similar to large Low Surface Brightness galaxies that contain massive dark matter haloes that are supposed to make discs stable against bar formation as predicted by classical theory (Hohl 1971; Ostriker & Peebles 1973). We see bar instability in our models despite the massive dark matter haloes; this is, similar to the observations of low surface brightness galaxies (e.g. UM 163; Honey et al. 2016). This can be explained by the live and rotating nature of haloes as seen in our study and others in the literature (Saha & Naab 2013; Long et al. 2014). In our case, we have assumed a constant halo spin value equal to 0.035.

Another important implication of this study is the continuous gain in angular momentum by the bulge. We see that for the models that do not form bars, there is no angular momentum exchange with the halo, as shown in Figs 13 and 14. But the bulge component always gains angular momentum irrespective of bar formation in all the models. The amount of angular momentum absorbed by the bulges is more in the case when the disc forms bars.

5 SUMMARY

In this paper, we have used *N*-body simulations to understand the effect of bulge mass and bulge concentration on bar formation and evolution in disc galaxies. We have evolved two types of models: MA models have dense bulges and discs of relatively high surface mass density Σ and MB models have less dense bulges and low Σ values compared to MA. We vary the bulge mass in similar steps for both models.

1. For models with concentrated bulges (MA), we notice that a bar forms earlier for bulgeless galaxies compared to those with bulges. The delay in bar formation increases as the bulge-to-disc mass fraction increases. No bar forms when the bulge-to-disc mass fraction is >0.3.

2. For models with less dense bulge (MB), the bar formation time-scale increases with bulge mass up to bulge-to-disc mass fraction equal to 0.5. No bar forms when the bulge-to-disc mass fractions is >0.6.

3. Our simulations show that bars are faster for models with massive bulges. The rate of decrease in pattern speed of the bar increases with introduction of massive bulges in our galaxy models.

4. We see that in both MA- and MB-type models, the introduction of more massive bulges makes the disc stellar dispersion higher in the inner disc region. Hence, massive bulges do not allow overall disc-halo angular momentum exchange that in turn makes the disc bar stable. The fraction of bulge-to-disc mass ratio required to make a disc bar stable varies with the density of bulge. In our study, we see that it has a value of 0.3 for the dense bulge models (MA) and 0.6 for the low-density bulge model (MB).

5. We put a limit on bar formation in discs with bulges by introducing a quantity B that is the ratio of radial force due to bulge and disc components ($B = F_{\mathrm{b}}/F_{\mathrm{tot}}$) at disc scale length ($R_{\mathrm{d}}$). We find that bars do not form for $B > 0.35$. This is because the velocity dispersion of disc stars becomes large enough to inhibit bar-type instability. This criterion does not depend on bulge concentration but takes into account the radial force due to a bulge that prohibits bar formation. It is similar to earlier values derived for discs with spherical haloes (Sellwood 1980).

6. The rate of angular momentum transfer to bulge and halo components from disc is more for dense bulges (MA) than less dense ones (MB).

7. Bulge component always gains total angular momentum of the same order irrespective of bar formation in any of the simulated models, while the gain in total angular momentum for halo component dampens by huge amount for bar stable models compared to bar unstable ones.

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